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WHEN SHOULD, AND SHOULDN'T, I USE LLMs TO SOLVE MATHEMATICS PROBLEMS?

Does using an LLM help me learn the mathematics, or does it do the thinking I need to practise?

BACK TO SCHOOL DAY
MATHEMATICS ROUNDTABLE

QUICK POLL
Stuck on homework. Would you ask AI?
Hands up: yes · no · it depends

THE DECISION BEFORE YOU PROMPT

USE IT

- For a hint when you're stuck
- To explain a concept
- For feedback on your own attempt
- For extra practice problems

ONLY IF YOUR PROFESSOR PERMITS IT

DON'T USE IT

- To produce work you must do independently
- During a prohibited assessment
- For an answer you cannot check

THE TEST Does it help me learn, or does it do the thinking I need to practise?

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CHOOSE YOUR TUTOR

RECOMMENDED

SOCRATIC TUTOR

- Asks one question at a time
- Gives small hints, not the answer
- Helps you find your own mistakes

Best while you are learning a method. You do the reasoning; it keeps you moving.

EXAM TUTOR

LATER

Once you know the method: practice questions one at a time, with feedback after each answer.

AI AS THE STUDENT

TO TEST YOURSELF

You explain the solution. The AI plays a learner and asks where your explanation is unclear.

PROMPT WITH CREATE

Each part of the prompt shapes how the AI tutors you through a math problem.

C	R	E	A	T	E
CHARACTER & CONTEXT	**REQUEST**	**EXAMPLES**	**ADDITIONAL INSTRUCTIONS**	**TYPE OF OUTPUT**	**EXTRAS**
I'm in Calculus I. I can differentiate polynomials, but I get lost with the chain rule when one function is inside another.	**Help me see where my integration by parts goes wrong in step 3.**	**Work through $x^2 + 7x + 12 = 0$ as an example, then let me try mine.**	**Only use methods from our course: no quadratic formula, no calculator. Don't correct my mistakes; ask a question that points to them.**	**Put each algebraic step on its own line, with a one-sentence reason.**	**Confirm the answer with the quadratic formula, then give me a similar problem to do on my own.**
sets who the AI is	sets the task	sets the size of help	sets the limits	sets the format	adds a final check

MONTY HALL PROBLEM

THE SETUP



One door hides a car; the other two hide goats. You pick Door 1. The host, who knows where the car is, opens another door to show a goat. Should you stay or switch?

VOTE: STAY OR SWITCH?
Hands up before the reveal.

TUTOR'S FIRST QUESTION
"If you always switch, in which cases do you lose?"

WHY SWITCHING WINS 2 TIMES IN 3

CAR IS BEHIND	STAY	SWITCH
Door 1 host opens 2 or 3	**Win**	Lose
Door 2 host must open 3	Lose	**Win**
Door 3 host must open 2	Lose	**Win**
CHANCE OF WINNING	**1 in 3**	**2 in 3**

CHANGE ONE RULE **If the host opens a door at random and it happens to show a goat, switching wins only 1 in 2. A memorised "2 in 3" would be wrong.**

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LIVE MATH EXAMPLE

THE PROBLEM

Solve $x^2 - 5x + 6 = 0$.

I'm stuck on how to factor it.

PROMPT A

"Solve this for me."

You get finished work. The factoring you need to practise is done for you.

PROMPT B · CREATE

C **Character & Context** I am learning to solve quadratic equations by factoring. Act as a Socratic math tutor.

R **Request** Help me solve $x^2 - 5x + 6 = 0$ through questions.

E **Examples** A useful hint would ask what two numbers multiply to 6 and add to −5.

A **Additional Instructions** Do not give me the factors or roots immediately. If I make a mistake, help me find it.

T **Type of Output** Ask one short question at a time and wait for my answer.

E **Extras** Once I have a solution, ask me to substitute both roots into the original equation.

YOUR TURN

Answer the tutor's question.

TUTOR'S FIRST REPLY

"Which two numbers multiply to 6 and add to −5?"

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CHECK THE RESULT

VERIFY EACH STEP YOURSELF

1 **FACTOR PAIR**
$(-2) \times (-3) = 6$ and $(-2) + (-3) = -5$ ✓

2 **FACTOR, THEN EXPAND TO CONFIRM**
$(x - 2)(x - 3) = x^2 - 3x - 2x + 6 = x^2 - 5x + 6$ ✓

3 **ZERO PRODUCT**
$x - 2 = 0$ or $x - 3 = 0$, so $x = 2$ or $x = 3$ ✓

4 **SUBSTITUTE BOTH ROOTS**
$x = 2$: $4 - 10 + 6 = 0$ $x = 3$: $9 - 15 + 6 = 0$ ✓

CHOOSE THE RIGHT CHECK

DIRECT CALCULATION

For a computation like this one: substitute, expand and test each step.

SIFT

When the LLM states a theorem, cites a source, or offers a method you cannot confirm yourself:

Stop · **I**nvestigate the source · **F**ind better coverage · **T**race claims to their original context

SO, DOES IT HELP ME LEARN? **Yes, when it guides and I do the thinking and the checking.**